UNIFORM CONDENSATION WINDOWS FOR ZERO-RADIUS RANDOM ALLOCATIONS

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Abstract. For any fixed nonnegative weight sequence with positive constant term and zero radius of convergence, we prove the existence of arbitrarily large intervals of total masses on which a single box carries almost all the mass, uniformly over the number of boxes at densities in a compact subset of $(0, \infty)$. Consequently every admissible full mass path of positive finite limiting density has a subsequence exhibiting complete condensation. This answers the zero-radius question in Janson's survey (Problem 19.41 in the published version, Problem 18.40 in the preprint). The argument preserves the small-box partition factor by an exact occupation-count insertion. A deterministic record-margin estimate then provides a uniform finite-size probability bound. We also show that multiplicative density of a sampled index set is necessary and sufficient for a universal subsequence-condensation statement.

1. Introduction

Weighted random allocations provide a common framework for occupancy models [6] and degree sequences of simply generated trees. The probability of an allocation is proportional to the product of the weights of its box sizes. When the total number of balls is proportional to the number of boxes, a basic question is whether one box carries a macroscopic fraction of the mass. For irregular weights this question is sensitive to the sizes at which the model is observed. Janson's survey [1] develops the allocation–tree correspondence and formulates several questions about such condensation. When the radius of convergence is positive and the density exceeds the mean ν of the tilted distribution, a fraction $\lambda - \nu$ of the mass escapes to boxes of unbounded size [1, Lemma 19.32]. For weights with power-law decay it collects in a single box; see Jonsson and Stefánsson [5] and Kortchemski [7] for the corresponding nongeneric trees, and Grosskinsky, Schütz and Spohn [4] for the zero-range process, whose stationary measures are allocations of this type.

We consider fixed weights $w_j \geq 0$ with $w_0 > 0$ for which the formal generating function $\Phi(z) = \sum_{j \geq 0} w_j z^j$ has radius zero. This is precisely the nontrivial $\nu = 0$ regime of the survey. No assumption is made on the ratios w_{j+1}/w_j or on the support. In particular, a large weight may be followed by an arbitrarily long interval of zero weights. A single probability law obtained by a fixed positive exponential tilt is unavailable: $\Phi(z) = \infty$ for every $z > 0$. We instead use finite truncations and record-dependent tilts, always retaining the original allocation partition function in the denominator.

The question addressed here is whether the largest box can be $o(n)$ in probability throughout a prescribed sequence $m_n/n \to \lambda > 0$. Janson's Problem 19.41 [1, p. 228], numbered Problem 18.40 in the preprint [2], asks this in the zero-radius regime. The answer given below is negative. More strongly, for every such path there is a subsequence along which a single box carries a fraction tending to one of the total mass. The main theorem is uniform over entire intervals of masses and over box counts in any fixed compact positive-density range. The intervals are chosen from the weights, rather than separately for each path. Arbitrary jumps and oscillations in m_n are permitted.

There are two obstructions to a direct comparison with an all-in-one-box configuration. First, the target mass need not belong to the support of the weights. Second, the total contribution of

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configurations consisting of a giant box and many small boxes need not be comparable to the weight of a single configuration. We resolve the first obstruction by a supported representation of the residual mass, and the second by an exact insertion inequality that keeps the whole small-box partition factor. Its factorial count compensates for the number of inserted large residual pieces. A geometric record decomposition bounds their weight loss. Summing that loss requires distinguishing scales at which the record increases from possibly long intervals on which it is constant.

The conclusion is genuinely subsequential. Sparse factorial support gives examples with a sublinear largest box on certain subsequences [1, Example 19.39]. The infinite-star local limit is already known in full generality: it holds for every zero-radius weight sequence with $w_0 > 0$ by [1, Theorem 7.1], and under a successive-weight-ratio hypothesis, with stronger size estimates for explicit factorial families, in Janson, Jonsson and Stefánsson [3]. Local convergence controls balls of a fixed radius, so it gives only $d^+(o) \to \infty$; Examples 19.37–19.39 of [1] satisfy its hypotheses while $Y_{(1)}/n$ oscillates, splits between two boxes, or vanishes along a subsequence. The statement proved here concerns the fraction of the mass and imposes only zero radius. We do not address the positive-boundary-mean question (Problem 19.40 of [1], Problem 18.39 of [2]).

We also identify the exact sampling condition for the universal statement. An increasing sampled index set admits a near-total-condensation subsequence for every zero-radius law and every admissible positive-density path if and only if the ratio of successive sampled indices tends to one. The converse is a deterministic support construction. For simply generated plane trees, the allocation theorem and the known size-biased root identity imply that, along a subsequence, the root has almost all vertices as children.

Organization. Section 2 states the main results. Sections 3 and 4 give the insertion inequality and the geometric summation bounds. Section 5 constructs supported residuals, and Section 6 gives a deterministic favorable-margin criterion. Section 7 proves the uniform theorem. Sections 8 and 9 treat paths, trees, and sampling sharpness. Section 10 records exact finite computations; none of these computations is used as a substitute for the proofs.

2. Model and results

Let $w = (w_j)_{j\ge 0}$ be fixed, with $w_0 > 0$, $0 \le w_j < \infty$, and

$$\rho = 0, \qquad \text{equivalently} \qquad \limsup_{j\to\infty} w_j^{1/j} = \infty. \tag{1}$$

For nonnegative integers m, n put

$$Z(m,n) = \sum_{y_1+\cdots+y_n=m} \prod_{v=1}^{n} w_{y_v}.$$

The pair (m,n) is *admissible* if $Z(m,n) > 0$. On such a pair,

$$\mathbb{P}_w(Y = y) = Z(m,n)^{-1} \prod_{v=1}^{n} w_{y_v}, \qquad Y_{(1)} \ge Y_{(2)} \ge \cdots$$

denote the ordered occupancies. The weights are independent of m, n.

Theorem 2.1 (Uniform condensation windows)**.** *Assume* (1)*. Fix* $\varepsilon \in (0,1)$ *and* $0 < \alpha < \beta < \infty$*. There are* $\delta > 0$ *and supported integers* $k_r \uparrow \infty$ *such that*

$$\sup_{\substack{(1+3\delta)k_r \le m \le (1+5\delta)k_r \\ \alpha \le m/n \le \beta,\ Z(m,n)>0}} \mathbb{P}_w\big(Y_{(1)}(m,n) \le (1-\varepsilon)m\big) \longrightarrow 0. \tag{2}$$

The supremum is over integer pairs. The windows contain admissible pairs for all sufficiently large r*; thus the conclusion is not vacuous.*

Corollary 2.2 (Arbitrary full paths and dense sampled paths)**.** *Let* $\mathcal{N} = \{N_1 < N_2 < \cdots\}$ *satisfy* $N_{j+1}/N_j \to 1$*; in particular one may take* $\mathcal{N} = \mathbb{N}$*. Suppose* (m_n, n) *is eventually admissible on*

$\mathcal{N}$ and $m_n/n \to \lambda \in (0,\infty)$. Then there are $n_r \in \mathcal{N}$, increasing to infinity, such that

$$\frac{Y_{(1)}(m_{n_r}, n_r)}{m_{n_r}} \xrightarrow{\mathbb{P}} 1. \tag{3}$$

Consequently

$$\limsup_{n\in\mathcal{N}} \mathbb{E}\frac{Y_{(1)}(m_n, n)}{n} = \lambda.$$

Along the selected subsequence $Y_{(1)}/n \to \lambda$ and $(m_n - Y_{(1)})/n \to 0$ both in probability and in L^1, and $Y_{(2)}/n \to 0$ in probability. For a full path, every fixed accuracy admits whole intervals of consecutive sizes with uniformly vanishing failure probability, as specified in Section 8.

The zero-radius question is Problem 19.41 of [1], or Problem 18.40 of [2]. It asks whether $Y_{(1)}/n \to 0$ in probability can hold along the full sequence when $\lambda > \nu = 0$. For nontrivial weights with $w_0 > 0$, this is the zero-radius regime. Corollary 2.2 rules it out without any restriction on the jumps or oscillations of m_n. This is a subsequence conclusion, not a claim of condensation at every size or of $\mathbb{P}(Y_{(1)} = m_n) \to 1$. The source already supplies sparse-support examples with sublinear maxima on selected subsequences.

The density condition on $\mathcal{N}$ in Corollary 2.2 is optimal for a statement valid for every weight sequence: Section 9 gives an explicit counterexample whenever that condition fails.

Normalize $w_0 = 1$ by dividing all weights by w_0, which leaves every allocation law unchanged. Let $d = \gcd\{j \geq 1 : w_j > 0\}$. Dividing supported sizes and admissible totals by d preserves $Y_{(1)}/m$ and zero radius, and replaces a density λ by λ/d. Until the proof of Theorem 2.1 is completed, we work with span one.

3. Insertion without losing the small-box factor

For an integer $u \geq 0$ write

$$Z_u(t,n) = \sum_{\substack{x_1+\cdots+x_n=t \\ 0\leq x_v\leq u}} \prod_{v=1}^{n} w_{x_v}.$$

The following exact finite inequality keeps the entire background partition factor. It is applied only to positive-weight background count vectors.

Lemma 3.1 (Occupation-count insertion)**.** *Let u, k, t, r, n and $K \geq 1$ be nonnegative integers, with $n \geq 1$, $u < k$, $w_0 = 1$, $w_k > 0$, and $m = t + k + r$. Suppose r has a fixed representation by R supported positive sizes at least K, together with C other supported positive sizes. All these sizes are less than k. Let W_r be the product of their weights. Put $Q = 1 + R + C$, and let r_j be the multiplicity of size j in the representation. If $t + Q \leq n$, then*

$$Z(m,n) \geq w_k W_r Z_u(t,n) \frac{(n-t)_{\underline{Q}}}{\displaystyle\prod_{j:r_j>0} (\min\{n, \lfloor m/j \rfloor\})_{\underline{r_j}}}. \tag{4}$$

In particular, if

$$nK \geq 2m, \qquad t + 1 + R + C \leq n/2, \tag{5}$$

then

$$Z(m,n) \geq \frac{n}{2^{C+1}} w_k W_r Z_u(t,n). \tag{6}$$

Repeated sizes, including sizes occurring in both parts of the representation, are allowed. An empty product equals one, and $(x)_{\underline{q}} = x(x-1)\cdots(x-q+1)$ for $q \geq 1$, with $(x)_{\underline{0}} = 1$. The normalization $w_0 = 1$ of Section 2 is in force throughout; for a general $w_0 > 0$ every right-hand side above carries an extra factor w_0^{-Q}, since the Q overwritten coordinates are empty in the background.

Proof. Describe a background configuration by its occupation counts a_j, where $\sum a_j = n$ and $\sum j a_j = t$, and $a_j = 0$ for $j > u$. The total weight of that count vector, including all coordinate permutations, is

$$\mathcal{W}(a) = \frac{n!}{\prod_j a_j!} \prod_j w_j^{a_j}.$$

Set $Q = 1+R+C$. Use Q empty coordinates to insert one size-k box and the chosen representation of r. If r_j is the multiplicity of size j in that representation, the resulting count vector is

$$b_0 = a_0 - Q, \quad b_k = 1, \quad b_j = a_j + r_j \quad (0 < j < k).$$

The map $a \mapsto b$ is injective: its inverse is $a_0 = b_0 + Q$, $a_j = b_j - r_j$ for $0 < j < k$, and $a_j = 0$ for $j \geq k$, the last because $u < k$. Its weight ratio is

$$\frac{\mathcal{W}(b)}{\mathcal{W}(a)} = w_k W_r \frac{(a_0)_{\underline{Q}}}{\prod_{0<j<k} \prod_{\ell=1}^{r_j} (a_j + \ell)}. \tag{7}$$

The jth denominator factor in (7) is $\prod_{\ell=1}^{r_j}(a_j+\ell) = (b_j)_{\underline{r_j}}$, and $x \mapsto (x)_{\underline{q}}$ is nondecreasing on integers $x \geq q$. Since $\sum_i b_i = n$ and $\sum_i i b_i = m$, the final count satisfies $b_j \leq \min\{n, \lfloor m/j \rfloor\}$, so that factor is at most $(\min\{n, \lfloor m/j \rfloor\})_{\underline{r_j}}$, a positive quantity. Bounding the numerator below by $(n-t)_{\underline{Q}}$, legitimate because $a_0 = n - \sum_{j\geq 1} a_j \geq n - \sum_{j\geq 1} j a_j = n - t \geq Q$, proves (4). For the simpler bound, the numerator is at least $(n/2)^Q$ by (5). Each denominator factor assigned to one of the R large pieces is at most m/K, since the final configuration has total m. Each of the other C factors is at most n. When the two classes share a size, assign their factors in either fixed order; both bounds still apply. Thus the ratio in (7) is at least

$$w_k W_r \frac{n}{2^{C+1}} \left(\frac{nK}{2m}\right)^R \geq w_k W_r \frac{n}{2^{C+1}}.$$

Summing over the distinct background count vectors gives (6). Weights of sizes not used in the construction only increase $Z(m,n)$. No assumption $m < 2k$ is required: every image count vector has exactly one size-k box by construction. □

The sharp inequality can also be obtained directly with labelled boxes. For every background, choose Q ordered distinct empty coordinates and insert the giant and a fixed ordered list of reservoir pieces. There are at least $(n-t)_{\underline{Q}}$ choices. A target has one possible giant location and at most $\prod_j (\min\{n, \lfloor m/j \rfloor\})_{\underline{r_j}}$ preimages of this kind. The weight multiplier is always $w_k W_r$, the Q overwritten coordinates being empty in the background and $w_0 = 1$. Weighted double counting gives (4) without passing through occupation classes.

4. Geometric records and two summation bounds

Fix a real base $a > 1$ and a constant $b > 2$. For every sufficiently large integer i define

$$B_i = \max_{\substack{1 \leq s \leq \lfloor a^i \rfloor \\ w_s > 0}} \frac{\log w_s + b \log s}{s}, \qquad s_i = \min \arg\max_{\substack{1 \leq s \leq \lfloor a^i \rfloor \\ w_s > 0}} \frac{\log w_s + b \log s}{s}, \qquad p_i = \frac{b \log s_i}{s_i}. \tag{8}$$

The feasible set in (8) grows with i, so B_i is nondecreasing. If $w_{j_\ell}^{1/j_\ell} \to \infty$ along a subsequence, then $B_i \geq j_\ell^{-1} \log w_{j_\ell}$ as soon as $\lfloor a^i \rfloor \geq j_\ell$, because $b \log j_\ell \geq 0$; hence $B_i \to \infty$. At a fixed s the quotient in (8) is a fixed finite number, so no s is maximizing for arbitrarily large i, and $s_i \to \infty$. Set $d_i = B_i - B_{i-1}$ and call an index *active* if $d_i > 0$. At an active index J,

$$a^{J-1} < s_J \leq a^J, \qquad \log w_{s_J} = s_J B_J - b \log s_J. \tag{9}$$

The lower bound holds because every index at most $\lfloor a^{J-1} \rfloor$ was already available at step $J-1$, so $s_J > \lfloor a^{J-1} \rfloor$; as s_J is an integer, $s_J \geq \lfloor a^{J-1} \rfloor + 1 > a^{J-1}$. By the definition of s_i as a maximizer, the identity $\log w_{s_i} = s_i B_i - b \log s_i$ holds at every index, active or not. Every index admissible at

step i but not at step $i-1$ exceeds $\lfloor a^{i-1} \rfloor \geq s_{i-1}$, so at an inactive index the smallest maximizing index remains unchanged. Consequently s_i and p_i are constant on each plateau of B_i.

Lemma 4.1 (Summable plateau charges)**.** *Fix a sufficiently large starting index i_0 and an integer $D \geq 1$. If $\mathcal{A}$ is the set of active indices after i_0, then*

$$\sum_{J \in \mathcal{A}} \sum_{i=i_0}^{J-D} a^{i-J} p_i < \infty. \tag{10}$$

More precisely, this sum is at most

$$\frac{a}{(a-1)^2} \left(p_{i_0} + \sum_{\ell \in \mathcal{A}} p_\ell \right),$$

and the parenthesized sum is finite.

Proof. For an active ℓ write $s_\ell = a^\ell / c$, where $c = a^\ell / s_\ell \in [1, a)$ by (9). Then $\log(s_\ell)/s_\ell = c(\ell \log a - \log c) a^{-\ell} \leq a\ell (\log a) a^{-\ell}$, so that

$$p_\ell \leq ba\ell (\log a) a^{-\ell},$$

which is summable. No lower bound on ℓ is needed. Consider a plateau starting at ℓ and ending just before the next active index v. A term from this plateau can contribute to (10) only at an active $J \geq v$: the starting active index $J = \ell$ has cutoff $J - D < \ell$. Its total contribution is at most

$$p_\ell \left(\sum_{i=\ell}^{v-1} a^i \right) \left(\sum_{J=v}^{\infty} a^{-J} \right) \leq \frac{a}{(a-1)^2} p_\ell.$$

The same bound applies to the initial plateau at i_0. There are always later active indices because $B_i \to \infty$. Summing proves the lemma. □

Lemma 4.2 (Contractive memory kernel)**.** *For $D \geq 2$ put*

$$C_a = 2a^2, \qquad \Theta(a, D) = 2a^{3-D} \left(\frac{D-1}{a-1} + \frac{1}{(a-1)^2} \right). \tag{11}$$

For every nondecreasing sequence B_i with the increments defined above, and every finite upper limit N,

$$C_a \sum_{\substack{i_0 < J \leq N \\ J \text{ active}}} \sum_{i=i_0}^{J-D} a^{i-J} (B_{J-1} - B_i) \leq \Theta(a, D) \sum_{\ell=i_0+1}^{N} d_\ell. \tag{12}$$

In particular D can be chosen so that $\Theta(a, D) < 1/4$.

Proof. Expand $B_{J-1} - B_i = \sum_{\ell=i+1}^{J-1} d_\ell$. All terms are nonnegative, so extend the sum over J to all larger integers and the lower limit on i to $-\infty$ to bound each coefficient. For a fixed ℓ and lag $t = J - \ell \geq 1$, that coefficient, before multiplication by C_a, is

$$\sum_{i \leq \min(J-D, \ell-1)} a^{i-J} = \begin{cases} a^{1-D}/(a-1), & 1 \leq t \leq D-1, \\ a^{-t}/(a-1), & t \geq D. \end{cases}$$

Summing over t gives $a^{1-D}((D-1)/(a-1) + 1/(a-1)^2)$. Multiplying by $2a^2$ gives (11). Exponential decay in D proves the last assertion. □

5. A supported representation of every small residual

The span-one reduction provides a finite set of positive supported sizes whose gcd is one. Its nonnegative additive semigroup contains every integer above some $h_* \geq 1$. Here is a direct justification. Choose one member s of the finite set. The nonnegative residues generated modulo s form a subgroup of the finite cyclic group and hence all residues. Choose one representable integer in each residue class; adding copies of s then represents every sufficiently large integer.

Fix an integer $K \geq 2$. Enlarge i_0 so that, for all $i \geq i_0$,

$$s_i \geq K, \qquad B_i \geq 0, \qquad a^{i_0} \geq h_*. \tag{13}$$

Choose once and for all a supported representation for each integer in

$$h_* \leq v < 2a^{i_0}.$$

Let C_* be a common bound on their numbers of positive parts, let $M = \lceil 2a^{i_0} \rceil$, and choose $C_0 \geq 0$ such that the logarithm of each of their product weights is at least $-C_0$. These are finite constants.

Lemma 5.1 (Greedy residual bound)**.** *Suppose $J \geq i_0 + D$, $r \geq h_*$, and*

$$I(r) := \lfloor \log_a(r/2) \rfloor \leq J - D, \qquad D \geq 2.$$

There is a representation of r using supported parts at least K, and at most C_* additional supported parts, with product weight W_r satisfying

$$B_{J-1} r - \log W_r \leq 2 \sum_{i=i_0}^{J-D} a^{i+1}(B_{J-1} - B_i + p_i) + M B_{J-1} + C_0. \tag{14}$$

All parts are at most r. The construction uses at most one stage at each geometric index i.

Proof. With current residual x, stop if $x < 2a^{i_0}$ and use its fixed terminal representation. Otherwise set

$$i = \lfloor \log_a(x/2) \rfloor, \quad q = \left\lfloor \frac{x - h_*}{s_i} \right\rfloor, \quad x' = x - q s_i.$$

Here $i \geq i_0$, $q \geq 1$, and

$$h_* \leq x' < h_* + s_i \leq 2a^i, \qquad q s_i \leq x < 2a^{i+1}.$$

Thus the next geometric index is strictly smaller than i. Add q copies of s_i and repeat. The procedure terminates in the fixed terminal range. At stage i the logarithmic weight per unit mass is $B_i - p_i$. Since $B_{J-1} \geq B_i$ for every index used, that stage's deficit against B_{J-1} is at most $2a^{i+1}(B_{J-1} - B_i + p_i)$. The terminal deficit is at most $M B_{J-1} + C_0$. Unused summands are nonnegative, giving (14). □

6. Deterministic margins

Fix $a > 1$, $b > 2$, and $D \geq 2$ with $\Theta(a, D) < 1/4$. Choose the finite terminal data i_0, M, C_0, C_* from the preceding section. For every active $J \geq i_0 + D$ set $k = s_J$, $h = B_{J-1}$, and define

$$E_J = 2 \sum_{i=i_0}^{J-D} a^{i+1}(B_{J-1} - B_i + p_i) + M B_{J-1} + C_0, \tag{15}$$

$$G_J = k(B_J - B_{J-1}) - b \log k - E_J. \tag{16}$$

All these quantities use finitely many weights. In particular G_J does not depend on a mass path or on a conditional allocation probability.

Lemma 6.1 (Unbounded favorable margins)**.** *For the fixed data above,*

$$\limsup_{\substack{J \to \infty \\ J \text{ active}}} G_J = +\infty. \tag{17}$$

Proof. Suppose instead $G_J \leq T$ for all sufficiently large active J. From (16), $k > a^{J-1}$ and $\log k \leq J \log a$, we obtain

$$\begin{aligned} d_J \leq C_a \sum_{i=i_0}^{J-D} a^{i-J}(B_{J-1} - B_i + p_i) \\ + M a^{1-J} B_{J-1} + a^{1-J}(bJ \log a + C_2), \end{aligned} \tag{18}$$

where $C_2 = \max\{0, C_0 + T\}$. Fix J_0 beyond this threshold with

$$\kappa = \frac{Ma^{1-J_0}}{a-1} < \frac{1}{4}. \tag{19}$$

For $N \geq J_0$, put $S_N = B_N - B_{J_0-1}$ and sum (18) over active $J \in [J_0, N]$. The memory term is at most $\Theta(a, D)S_N$ plus a fixed finite-prefix contribution, by Lemma 4.2. All p_i terms have a finite total, by Lemma 4.1; the last term is also summable. The terminal term satisfies

$$\begin{aligned} M \sum_{J=J_0}^{N} a^{1-J} B_{J-1} &\leq \frac{MB_{J_0-1}a^{2-J_0}}{a-1} + M \sum_{\ell=J_0}^{N-1} d_\ell \sum_{J=\ell+1}^{\infty} a^{1-J} \\ &\leq C_3 + \kappa S_N, \qquad C_3 = \frac{MB_{J_0-1}a^{2-J_0}}{a-1}. \end{aligned}$$

Thus $S_N \leq (\Theta(a, D) + \kappa)S_N + C_4$, where C_4 collects C_3, the fixed finite-prefix contribution of Lemma 4.2, the total supplied by Lemma 4.1, and the summable last term of (18). All four are independent of N. The coefficient is below $1/2$, contradicting $B_N \to \infty$. □

The preceding proof yields a finite-prefix version, without assuming any bound on G_J. It will also make the cancellation of all errors explicit. For $L \geq 0$ define

$$T_L = \sum_{j=L}^{\infty} a^{1-j} = \frac{a^{2-L}}{a-1}, \qquad U_L = \sum_{j=L}^{\infty} j a^{1-j} = a^{2-L}\left(\frac{L}{a-1} + \frac{1}{(a-1)^2}\right).$$

Set

$$\Pi = \frac{a}{(a-1)^2}\left(p_{i_0} + ba \log a \sum_{\ell=i_0+1}^{\infty} \ell a^{-\ell}\right)$$

and, for $J_0 \geq i_0 + D$ with (19), set

$$\begin{aligned} \mathcal{C} = \Theta(a, D)(B_{J_0-1} - B_{i_0}) + C_a\Pi + MB_{J_0-1}T_{J_0} \\ + b(\log a)U_{J_0} + C_0T_{J_0}. \end{aligned} \tag{20}$$

These are finite nonnegative constants. The two infinite sums have displayed geometric closed forms (the sum in Π equals U_{i_0+1}/a).

Lemma 6.2 (Finite-prefix margin bound)**.** *For all $N \geq J_0$,*

$$\sum_{\substack{J_0 \leq J \leq N \\ J \text{ active}}} \frac{G_J}{s_J} \geq (1 - \Theta(a, D) - \kappa)(B_N - B_{J_0-1}) - \mathcal{C}. \tag{21}$$

In particular, with $x_+ = \max\{x, 0\}$ and the maximum of an empty set interpreted as zero,

$$\max_{\substack{J_0 \leq J \leq N \\ J \text{ active}}} (G_J)_+ \geq \frac{(1 - \Theta(a, D) - \kappa)(B_N - B_{J_0-1}) - \mathcal{C}}{T_{J_0}}. \tag{22}$$

Proof. Use the identity $d_J = G_J/s_J + b\log(s_J)/s_J + E_J/s_J$ and sum over active indices. Each E_J is nonnegative, by $B_{J-1} \geq B_i$, $p_i \geq 0$, $B_{J-1} \geq 0$ and $C_0 \geq 0$, so $E_J/s_J \leq a^{1-J}E_J$ by (9). The memory term and the terminal term have the bounds from the preceding proof; none of those bounds involves T_{J_0}, so the two lemmas are independent, and Lemma 6.1 also follows from (21) together with $B_N \to \infty$. Lemma 4.1 and $p_\ell \leq ba\ell(\log a)a^{-\ell}$ at active ℓ bound the penalty term by $C_a\Pi$. Finally,

$$\sum_{J \geq J_0} \frac{b\log s_J + C_0}{s_J} \leq b(\log a)U_{J_0} + C_0T_{J_0},$$

where sums on the left are restricted to active indices. Rearrangement gives (21). Since $\sum_{J \geq J_0,\, J \text{ active}} s_J^{-1} \leq T_{J_0}$, bounding each G_J by the maximum of their positive parts gives (22). □

7. Uniform windows

Fix $\varepsilon \in (0,1)$ and $0 < \alpha < \beta < \infty$. Choose $a > 1$ close to one, fix $b = 4$, and choose $D \geq 2$ with $\Theta(a, D) < 1/4$. Next choose $\delta > 0$ such that

$$\delta < \min\left\{\frac{a^{-D}}{100}, \frac{1}{100(1+\beta)}, \frac{1}{100}\right\}, \qquad \frac{1}{a(1+5\delta)} > 1 - \varepsilon. \tag{23}$$

Choose an integer $K \geq 4(1+\beta)$, and then the fixed terminal data from Section 5. Every parameter is now fixed, before letting $J \to \infty$. At an active J write $k = s_J$, $u = \lfloor a^{J-1} \rfloor$, $h = B_{J-1}$, and define the finite auxiliary distribution

$$q_j = w_j e^{-hj}, \quad A_J = \sum_{j=0}^{u} q_j, \quad p_J(j) = q_j/A_J, \quad \mu_J = \sum_{j=0}^{u} j p_J(j). \tag{24}$$

The normalization $w_0 = 1$ gives $q_0 = 1$, and the record definition implies $q_j \leq j^{-b}$ for $1 \leq j \leq u$. Since $h \to \infty$, dominated convergence gives $A_J \to 1$ and $\mu_J \to 0$, using $\sum j^{1-b} < \infty$ for the latter assertion.

Lemma 7.1 (Uniform finite-size certificate)**.** *For all sufficiently large active J, every integer pair with*

$$(1+3\delta)k \leq m \leq (1+5\delta)k, \qquad \alpha \leq m/n \leq \beta \tag{25}$$

is admissible in the span-one model, and

$$\mathbb{P}_w(Y_{(1)}(m,n) \leq u) \leq \min\{1, Ce^{-G_J}\}, \qquad C = \max\left\{1, \frac{2^{C_*+3}\beta\delta}{1+3\delta}\right\}. \tag{26}$$

Here G_J is the deterministic quantity in (16)*; the same bound holds for every pair in the window.*

Proof. The sufficiently large condition can be checked using only finite data: require $J \geq i_0 + D$, $\delta k \geq 1$, $2\delta k \geq h_*$,

$$\mu_J \leq \frac{\alpha\delta}{2(1+5\delta)}, \qquad 5\delta k + 1 \leq \frac{(1+3\delta)k}{2\beta}, \tag{27}$$

and $h \geq 0$. These all hold eventually. For the sum S_n of n independent copies of (24), Markov's inequality gives

$$\mathbb{P}(S_n > \delta k) \leq \frac{\mathbb{E}S_n}{\delta k} = \frac{n\mu_J}{\delta k} \leq \frac{(1+5\delta)\mu_J}{\alpha\delta} \leq \frac{1}{2},$$

using $n \leq (1+5\delta)k/\alpha$ and then (27). Hence $\mathbb{P}(S_n \leq \delta k) \geq 1/2$ uniformly in (25), and some integer $t \in [0, \lfloor \delta k \rfloor]$ satisfies

$$\mathbb{P}(S_n = t) \geq \frac{1}{2(\lfloor \delta k \rfloor + 1)}. \tag{28}$$

Put $r = m - k - t$, so $2\delta k \leq r \leq 5\delta k$. By (23), $I(r) \leq J - D$. Lemma 5.1 supplies a representation with product W_r and deficit $hr - \log W_r \leq E_J$. If it uses R large pieces and $C' \leq C_*$ terminal pieces, then

$$t + 1 + R + C' \leq t + 1 + r = m - k + 1 \leq 5\delta k + 1 \leq n/2.$$

Also $nK \geq 2m$, and every inserted part is smaller than k, since $r \leq 5\delta k < k$. Lemma 3.1 gives

$$Z(m,n) \geq \frac{n}{2^{C_*+1}} w_k W_r Z_u(t,n) > 0.$$

Thus no admissibility assumption has been used in producing the bound. The exact identities and inequality

$$Z_u(t,n) = e^{ht} A_J^n \mathbb{P}(S_n = t), \qquad Z_u(m,n) = e^{hm} A_J^n \mathbb{P}(S_n = m) \leq e^{hm} A_J^n$$

now yield

$$\mathbb{P}_w(Y_{(1)} \le u) \le \frac{2^{C_*+2}(\lfloor \delta k \rfloor + 1)}{n} \exp\{hk - \log w_k + hr - \log W_r\}$$
$$\le \frac{2^{C_*+3}\beta\delta}{1+3\delta}\, e^{-G_J}.$$

The whole small-box factor A_J^n cancels exactly. The last line uses $\delta k \ge 1$, $n \ge (1+3\delta)k/\beta$, and $\log w_k = kB_J - b\log k$. This proves (26). □

Proof of Theorem 2.1. Lemma 6.1 supplies active $J_r \uparrow \infty$ with $G_{J_r} \to \infty$. Let $k_r = s_{J_r}$. For all pairs in the corresponding window,

$$\frac{u}{m} \ge \frac{a^{J_r-1}-1}{(1+5\delta)a^{J_r}} \longrightarrow \frac{1}{a(1+5\delta)} > 1-\varepsilon \quad \text{as a lower bound.}$$

Thus (26) proves (2). The integer windows have positive linear length, and $\alpha < \beta$ guarantees that for every sufficiently large integer m in such a window there is an integer n with $\alpha \le m/n \le \beta$. Admissibility follows from the certificate. For span d, apply the argument to $w'_j = w_{dj}$, which again has $w'_0 = 1$, span one and zero radius, with masses m/d and the density interval $[\alpha/d, \beta/d]$; the map $y \mapsto y/d$ is a weight-preserving bijection of allocations and leaves $Y_{(1)}/m$ unchanged. Writing $k_r = dk'_r$, which is supported for w, the window $(1+3\delta)k'_r \le m/d \le (1+5\delta)k'_r$ is exactly $(1+3\delta)k_r \le m \le (1+5\delta)k_r$. Integers m in that window which are not multiples of d have $Z(m,n) = 0$ and are excluded by the constraint in (2); multiples of d do occur there because the window has length $2\delta k_r \to \infty$, and for each of them the span-one certificate supplies an admissible n. □

8. Consequences for full paths

Proof of Corollary 2.2. Fix $\varepsilon > 0$ and apply Theorem 2.1 with the density interval $[\lambda/2, 2\lambda]$. For each resulting k_r, let n_r be the first element of $\mathcal{N}$ at least $(1+4\delta)k_r/\lambda$. The condition $N_{j+1}/N_j \to 1$ implies

$$\frac{n_r}{k_r} \to \frac{1+4\delta}{\lambda}, \qquad \frac{m_{n_r}}{k_r} \to 1+4\delta.$$

Thus these pairs eventually lie in the windows, and their failure probabilities at accuracy ε tend to zero. The parameters δ, k_r and n_r produced here all depend on ε. Apply the assertion separately for each $i \ge 1$ with $\varepsilon = 1/i$, obtaining sizes $n_1^{(i)} < n_2^{(i)} < \cdots$ in $\mathcal{N}$ along which $\mathbb{P}_w(Y_{(1)} \le (1-1/i)m_n) \to 0$. Choose indices $\rho(i)$ recursively so that $\tilde{n}_i := n_{\rho(i)}^{(i)} > \tilde{n}_{i-1}$ and $\mathbb{P}_w\big(Y_{(1)}(m_{\tilde{n}_i}, \tilde{n}_i) \le (1-1/i)m_{\tilde{n}_i}\big) < 1/i$; both are possible because $n_\rho^{(i)} \to \infty$ in ρ and the displayed limit is zero. For a fixed $\varepsilon > 0$ and every $i \ge 1/\varepsilon$ the event $\{Y_{(1)} \le (1-\varepsilon)m\}$ is contained in $\{Y_{(1)} \le (1-1/i)m\}$, so the corresponding probability is below $1/i$. This proves (3) along $(\tilde{n}_i)$. The bounds $0 \le Y_{(1)}/m_n \le 1$ and $m_n/n \to \lambda$ give the expectation and L^1 assertions. Finally $Y_{(2)} \le m_n - Y_{(1)}$. □

For a full path $\mathcal{N} = \mathbb{N}$, the conclusion for a fixed accuracy is stronger than the selection of one size per record. Fix $\varepsilon > 0$ and let δ and k_r be as produced by Theorem 2.1 at that accuracy and at the density interval $[\lambda/2, 2\lambda]$. Define

$$I_r = \left[\left\lceil \frac{(1+\frac{7}{2}\delta)k_r}{\lambda} \right\rceil, \left\lfloor \frac{(1+\frac{9}{2}\delta)k_r}{\lambda} \right\rfloor\right] \cap \mathbb{N}. \tag{29}$$

Uniformly for $n \in I_r$, the relation $m_n - \lambda n = o(n)$ implies $|m_n - \lambda n| = o(k_r)$. Hence all such pairs eventually lie in (25), and

$$\sup_{n \in I_r} \mathbb{P}_w(Y_{(1)}(m_n, n) \le (1-\varepsilon)m_n) \to 0, \qquad |I_r| \sim \frac{\delta k_r}{\lambda}. \tag{30}$$

In particular the intervals have a fixed positive relative width at every fixed accuracy. They need not occur at regularly spaced locations. No rate uniform in the weight sequence is asserted.

For simply generated plane trees the allocation has $m = n - 1$. The cyclic lemma groups every allocation sequence of total $n - 1$ into an orbit of n rotations containing exactly one valid tree sequence; product weights and the maximum are unchanged. That orbit has exactly n elements: a period $p \mid n$ with $p < n$ would force $g = n/p \geq 2$ to divide both n and the total $n - 1$, hence $g = 1$. Thus the maximum outdegree has the same law as $Y_{(1)}$. Admissible sufficiently large tree sizes form the lattice $n \equiv 1 \pmod{d}$: to verify eventual admissibility, choose a finite subset of supported degrees of gcd d, represent every sufficiently large multiple of d as their nonnegative sum, and pad with zeros to n entries. There are at most $n - 1$ positive summands since every degree is at least one. The dense-index extension therefore applies.

Corollary 8.1 (Subsequence concentration at the root)**.** *For simply generated plane trees with zero-radius weights and $w_0 > 0$, there are admissible $n_r \uparrow \infty$ such that the root has $(1 - o_{\mathbb{P}}(1))n_r$ children. With probability tending to one the root is the unique vertex of maximum outdegree. The number of vertices at depth at least two is $o_{\mathbb{P}}(n_r)$, also in expectation after division by n_r.*

Proof. Fix $n \geq 2$. Let A_j be the number of vertices of outdegree j, and let $D = \max\{j : A_j > 0\}$. Janson's root-degree identity [2, Lemma 14.7], Lemma 15.7 of [1], states that $d^+(o)$ has the size-biased law of a single occupancy. Its conditional form is

$$\mathbb{P}(d^+(o) = j \mid (A_i)) = \frac{jA_j}{n-1}, \tag{31}$$

which the following count establishes directly. The number of plane trees with these degree counts is $(n-1)!/\prod_i A_i!$. Removing a root of degree j leaves an ordered forest of j trees. The cyclic lemma counts these forests as $j(n-2)!/((A_j - 1)!\prod_{i \neq j} A_i!)$, yielding (31). All trees with given counts have the same product weight, so conditioning on the counts does not alter this calculation.

Along the subsequence from Corollary 2.2, $D/(n-1) \to 1$ in probability. On $D > (n-1)/2$ the maximum is unique, and its probability of being the root, conditionally on the counts, is $D/(n-1)$. Taking expectations gives probability tending to one: explicitly, the error is at most $\mathbb{P}(D \leq (n-1)/2) + \mathbb{E}[1 - D/(n-1)]$, which tends to zero. This also proves $d^+(o)/(n-1) \to 1$ in probability and in L^1. Finally the number of vertices at depth at least two is exactly $n - 1 - d^+(o)$. □

The root identity is not a new result, and neither is the infinite-star local limit: that holds for every zero-radius weight sequence with $w_0 > 0$ by [1, Theorem 7.1], and under a successive-weight-ratio hypothesis in [3]. Local convergence controls balls of a fixed radius and therefore gives only $d^+(o) \to \infty$. The corollary above concerns the fraction of the mass, and is obtained from the unrestricted subsequence statement proved here.

9. The sampling condition is sharp

Theorem 9.1 (Sharp universal sampling criterion)**.** *For an infinite index set $\mathcal{N} = \{N_1 < N_2 < \cdots\}$, the following are equivalent:*

(i) $N_{j+1}/N_j \to 1$.

(ii) For every fixed zero-radius weight sequence with $w_0 > 0$ and every eventually admissible mass path on $\mathcal{N}$ with positive finite limiting density, some subsequence satisfies $Y_{(1)}/m \to 1$ in probability.

If $\limsup_j N_{j+1}/N_j = \infty$, there are such weights, with $w_1 > 0$, for which $Y_{(1)}(n-1, n)/n \to 0$ deterministically throughout $n \in \mathcal{N}$.

Proof. The implication (i)⇒(ii) is Corollary 2.2. If (i) fails, there exist $\theta > 1$ and infinitely many increasing indices r_j with $N_{r_j+1} \geq \theta N_{r_j}$. Discard finitely many so that $N_{r_j} \geq 2$. Set

$$S = \{0, 1\} \cup \{N_{r_j} : j \geq 1\}, \qquad w_0 = w_1 = 1, \quad w_k = e^{k^2} \ (k \in S \setminus \{0, 1\}),$$

and set every other weight to zero. These are fixed weights with zero radius. The tree path $m_n = n - 1$ is admissible at every size by using ones and zeros. If a supported $k = N_{r_j}$ is available

at a sampled size $n \in \mathcal{N}$, then $k \le n-1$, so $n \ge N_{r_j+1} \ge \theta k$. Consequently

$$Y_{(1)}(n-1,n) \le \max\{1, n/\theta\} \quad (n \in \mathcal{N})$$

for every configuration. This implies $\limsup_{n\in\mathcal{N}} Y_{(1)}/(n-1) \le 1/\theta < 1$, disproving (ii).

For the final assertion, select r_j so that $N_{r_j+1}/N_{r_j} \ge j+1$ and use the same construction. Fix L. Any available supported N_{r_j} with $j \ge L$ is at most $n/(L+1)$. The finitely many earlier supported values, divided by n, tend to zero. Thus $\limsup_{n\in\mathcal{N}} Y_{(1)}/n \le 1/(L+1)$ deterministically. Let $L \to \infty$. □

This is an optimality statement about a universal theorem, not an assertion that every weight sequence fails on a nondense set. For example, the powers of two as sampled sizes and as nonzero large support already force a maximum at most $n/2$ on the tree path; factorial gaps can force a vanishing fraction, in the same spirit as Example 19.39 of [1].

10. Exact finite illustrations

The argument uses a deterministic margin to give a uniform window statement and a sharp sampling criterion. The finite checks below test its algebraic components; they are not a proof-assistant formalization or an independent audit of the limiting argument.

The accompanying `code/check_uniform_windows.py` evaluates the exact failure probability and both insertion bounds on an explicit window of integer pairs with an empty background ($t = 0$), the exact finite-prefix inequality with rational geometric data, and the support obstructions at powers-of-two and factorial sample sizes. The insertion inequality with a nontrivial background, including totals $m \ge 2k$, is exercised in `code/check_full_path.py` and `code/audit_stress_checks.py`. The window parameters below are chosen for legibility and do not satisfy (23). In that window the weights are supported on $\{0, 2, 3, 5, 17\}$ below 120, with $w_j = 2^{j^2}$ at the positive supported values; assigning the same weights at $j = \ell!$ for $\ell \ge 5$ makes this one fixed infinite zero-radius law. Every pair $20 \le m \le 22$, $1/2 \le m/n \le 2$ is checked with exact arithmetic. There are 97 such pairs. The sharp insertion bound certifies

$$\sup_{\substack{20\le m\le 22\\ 1/2\le m/n\le 2}} \mathbb{P}_w(Y_{(1)} \le 3m/4) \le 2^{-189}.$$

The comparison is summarized in Table 1. All probabilities and bounds are computed as rational numbers; decimal values are displays, not the basis of the assertions. The exact rational values

Table 1. An exact finite illustration over all 97 admissible pairs $20 \le m \le 22$, $1/2 \le m/n \le 2$.

Quantity	Maximum over the window
Actual failure probability	$1.50986220045 \cdot 10^{-58}$
Sharp insertion upper bound	$9.05917320269 \cdot 10^{-58} \le 2^{-189}$
Coarse insertion upper bound	$7.58157669219 \cdot 10^{-56} \le 2^{-183}$

are saved in the checker output. The checker also verifies the root-degree identity by enumerating all plane trees of sizes two through ten. These finite computations do not replace the asymptotic argument.

Ten of the finite algebraic steps are additionally machine-checked in Lean 4 against Mathlib, in `lean/`: the falling-factorial numerator bound and the exponent bookkeeping in the proof of Lemma 3.1; the closed forms T_L and U_L; the coefficient total behind $\Theta(a, D)$ in Lemma 4.2 and its identity; the bound $q_j \le j^{-b}$ of (24); the Markov threshold of (27); and the strict decrease of the geometric index in Lemma 5.1. Each compiles with no axioms beyond `propext`, `Classical.choice` and `Quot.sound`. The asymptotic argument is not formalized.

The positive-boundary-mean question is outside the scope of the present results. The factorial and successive-ratio results in [3], and the size-biased root identity, are used only with their stated attribution.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

This manuscript was developed with AI assistance as part of an ongoing AI-assisted mathematical research project, provisionally called Math Brain. OpenAI's ChatGPT contributed to the mathematical arguments, literature searches, computations and writing, and performed five documented review passes of the resulting draft. That draft was then examined by three independent audit passes of Anthropic's Claude Opus 5, covering the finite combinatorial core, the record and margin estimates, and the probabilistic assembly together with the tree and sampling consequences and the attribution of the problem and of prior work; the present version incorporates their corrections. The Lean development described in Section 10 was produced with the same assistance. None of this is independent referee review, human peer review, or a formal verification of the asymptotic theorem. The author directed the project and is responsible for decisions concerning dissemination. The present version is a research draft for further mathematical review.

Data and code availability

The complete LaTeX source, the exact-arithmetic Python scripts and their saved outputs, the Lean 4 files, and the review reports are available at `https://github.com/yspennstate/uniform-condensation-windows`. The Python scripts use only the standard library; the Lean files build against Mathlib. No external empirical dataset is required. The auxiliary computations verify finite identities and inequalities; they do not establish the asymptotic theorem by exhaustive testing.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [8].

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